

Who Benefits from Scotland's Alcohol Minimum Unit Pricing?

A Novel CITS-MAIHDA Framework for Intersectional Inequality Evaluation



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THE PROBLEM

BACKGROUND / CONTEXT

Scotland's Minimum Unit Pricing (MUP) targets heavy drinkers by raising the national floor price of alcohol. Overall consumption reduced, but the **inequalities of benefit** across **population subgroups** remains poorly understood.

THE GAP IN EVIDENCE

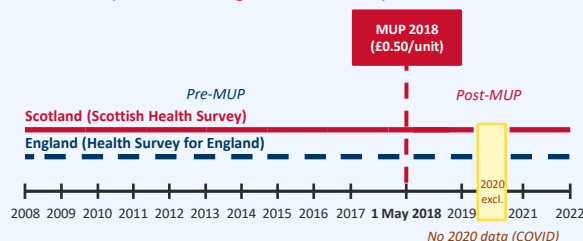
Current MUP evaluations show contradictory results as they examine **age, sex, or deprivation** in isolation. No study has holistically mapped how these **dimensions intersect** to shape both **additive and interaction-driven** inequalities.

RESEARCH AIMS

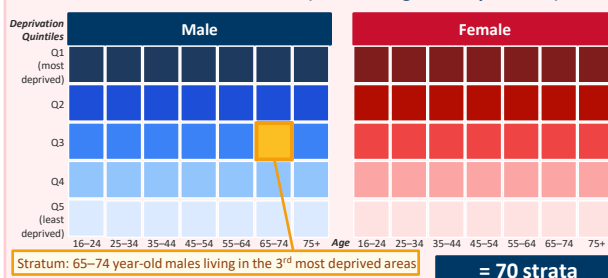
- ① Analyse historical trend in alcohol consumption
- ② Evaluate overall MUP impact on consumption.
- ③ Examine unequal MUP impacts via both isolated and intersectional lens (age × sex × deprivation)

THE FRAMEWORK

CITS DESIGN (Scotland vs England, 2008–2022)



MAIHDA: 70 Intersectional Strata (2 Sex × 7 Age × 5 Deprivation)



STRENGTH & DATA

METHODOLOGICAL STRENGTH

- Integrate intersectional MAIHDA (multilevel analysis of individual heterogeneity and discriminatory accuracy) model into CITS design for causal inference
- Derive real-world interaction effect from contrasting Bayesian posterior distributions

DATA & DESIGN

- 👤 National Cohorts: Scottish population vs English population
- 📅 Timeline: Annual repeated cross-sectional surveys from 2008 to 2022
- ⚙️ Modelling: Bayesian multilevel MCMC estimation via "brms" (R package)

Chapter 1 (Current Stage): Alcohol consumption

Chapter 2: Alcohol-specific hospitalisations

Chapter 3: Alcohol-specific mortality

TAKE-AWAY MESSAGE:

MUP reduced drinking overall — but did it reach those who need it most? We are building a novel framework to unmask who benefits, and who is left behind.

Downloadable version & references



References

BACKGROUND / CONTEXT

- ANDERSON, P., O'DONNELL, A., KANER, E. F. S., JANÉ-LLOPIS, E., MANTHEY, J. & REHM, J. 2021. Impact of minimum unit pricing on alcohol purchases in Scotland and Wales: controlled interrupted time series analyses. *The Lancet Public Health*, 6, **e557-e565**.
- GILES, L., MACKAY, D., RICHARDSON, E., LEWSEY, J., ROBINSON, M. & BEESTON, C. 2024. Evaluating the impact of minimum unit pricing (MUP) on alcohol sales after 3 years of implementation in Scotland: A controlled interrupted time-series study. *Addiction*, 119, **1378-1386**.

THE GAP IN EVIDENCE

- REHM, J., O'DONNELL, A., KANER, E. F. S., JANE, L. E., MANTHEY, J. & ANDERSON, P. 2022. Differential impact of minimum unit pricing on alcohol consumption between Scottish men and women: controlled interrupted time series analysis. *BMJ Open*, 12, **e054161**.
- MOHAN, G. 2025. An evaluation of the impact of a national Minimum Unit Price on alcohol policy on alcohol behaviours. *Journal of Public Health*, 47, **e94-e105**.

METHODOLOGICAL STRENGTH

- EVANS, C. R., LECKIE, G., SUBRAMANIAN, S. V., BELL, A. & MERLO, J. 2024. A tutorial for conducting intersectional multilevel analysis of individual heterogeneity and discriminatory accuracy (MAIHDA). *SSM - Population Health*, 26, **101664**.
- BAUER, G. R., CHURCHILL, S. M., MAHENDRAN, M., WALWYN, C., LIZOTTE, D. & VILLA-RUEDA, A. A. 2021. Intersectionality in quantitative research: A systematic review of its emergence and applications of theory and methods. *SSM - Population Health*, 14, **100798**.

DATA & DESIGN

- BÜRKNER, P.-C. 2017. brms: An R Package for Bayesian Multilevel Models Using Stan. *Journal of Statistical Software*, 80, **1 - 28**.
- ScotCen Social Research. (2019). *Scottish Health Survey*. [data series]. 2nd Release. UK Data Service. SN: 2000047, [DOI: http://doi.org/10.5255/UKDA-Series-2000047](http://doi.org/10.5255/UKDA-Series-2000047)
- NatCen Social Research, University College London, Department of Epidemiology and Public Health. (2023). *Health Survey for England*. [data series]. 7th Release. UK Data Service. SN: 2000021, [DOI: http://doi.org/10.5255/UKDA-Series-2000021](http://doi.org/10.5255/UKDA-Series-2000021)